# Aszad Hossain Akib

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**Superheavy Element Chemistry | Surface Adsorption | Atomistic Simulation | Machine-Learned Interatomic Potentials | Computational Materials Science**

## Career Objectives:

To obtain a fully funded graduate research internship in nanotechnology and materials science, leveraging my academic background, research skills, and passion for synthesizing nanoparticles in a sustainable way.

## Educational Background

**Master of Science in Computational Food Engineering (pursuing) | Thesis-based | 1st July, 2025 – 1st July, 2026 (Exp.)**

Department of Physical and Mathematical Sciences | Chattogram Veterinary and Animal Sciences University, Chattogram Bangladesh

First semester: 3.83/4.00

Thesis title: *Fabrication of dye-sensitized TiO2 sensors for gas sensing application.*

**Bachelor of Science in Food Science and Technology | 1st January, 2020 - 24th May, 2025**

Faculty of Food Science and Technology | Chattogram Veterinary and Animal Sciences University, Chattogram Bangladesh

Result: 3.46/4.00

Final year: 3.60/4.00

## Awards:

- **August, 2024 | New Product Development**
  Secured Bronze in New Product Development among 8 teams in **the UMT Mobility Program**.
- **June, 2024 | Internship grant $500**
  Provided by Chattogram Veterinary and Animal Sciences University to help complete all the internships along with the externship program at Universiti Malaysia Terengganu.
- **March, 2025 | UGC Grant | Research grant $350**

Offered by the University Grants Commission, Bangladesh for a research proposal on thin film materials for antibacterial surfaces.

- **March, 2026 | NST Fellowship | Research grant $450**
  Offered the prestigious NST Fellowship by the Ministry of Science and Technology for a research proposal on sustainable gas sensors.
- **July, 2026 | Partial Funded | Beihang International Summer School**
  Offered free registration, accommodation, meals, and partial travel expense coverage for attending the prestigious Beihang International Summer School
- **August,2026 | Full Fee Waiver ($150) | Online Participation**
  Granted Full registration fee waiver to attend the program MemrisTec Summer School 2026 for writing an article for the 90th birthday of Leon Chua.

## Teaching & Mentoring Experience:

- **Science Tutor | Private Tutor | 2022-2025**
  1. Taught 10 SSC/HSC students in Math, Biology, Chemistry, Physics, covering theory and practical lab work
  2. Helped the Students achieve a GPA of 5.00 out of 5.00
- **Introductory Research Advisor | 2022-2025**
  1. Guided and mentored 20 individuals across different projects, familiarizing them about research methodology concepts, basic descriptive statistics.
  2. Held meetings, feedback sessions, and delivered teaching materials that helped team members progress through the research project.

## Work Experience:

- **2025-Ongoing | Research Assistant (Volunteering) to Dr. Indrajit Saha | Chattogram Veterinary and Animal Sciences University | Department of Physical and Mathematical Sciences**
  Developed thin films using the Spray Pyrolysis and Spin Coating Process for determining the sensing properties (Glucose and Gas sensing for food spoilage) of thin films.
  Current Project: Assessing the antibacterial, water purification, and gas-sensing properties of metal-doped $TiO_2$, ZnO, and $Co_3O_4$.

## Research Experience:

- **July,2025 - Ongoing | Position not known | ABCD Laboratory, Bangladesh (since July 2025)**
  Joined the Advanced Bioinformatics, Computational Biology, and Data Science Laboratory, Bangladesh (ABCD Laboratory, Bangladesh) and am actively working on projects under the Supervision of Mr. Junaid.

- **Oct, 2025 - Jan, 2026 | Research Assistant (Volunteering Position) | BCSIR, Chattogram (Since October, 2025)**
  Actively working on projects related to determining antioxidant properties of LAB and also isolating bacteriophage from different samples.
- **July, 2024 – july2025 | Junior Research Assistant (Volunteering position) | Bangladesh Institute of Innovative Health Research - BIIHR (since July 2024)**
  Joined Bangladesh Institute of Innovative Health Research (BIIHR) as a Research Assistant on 1st July 2024. Actively working on projects under the Department of Health Sciences and Informatics

## Research Skills:

- **Experimental skills:**

  Natural Products Chemistry, Product Development, Proximate Analysis, Extraction, Characterization, Isolation.

- **Thin film Characterization:**

  Transmittance, Reflectance, Absorbance, Absorption Coefficient, Refractive index, Band Gap Determination, Urbach Energy, ISO 15184, ISO 2409, Contact Angle Measurement, Surface Energy Calculation, Antifogging, Self- cleaning/Dirt Removal, Refractive index measurement, ISO 27447, ISO 22196.

- **Food analysis skills:**

  Food Quality Analysis, Food Safety Assessment, Sensory Evaluation, Microbial Analysis, Fermentation Techniques, Enzyme Activity Assays, Food Additive Testing, Antioxidant Activity Assays, Shelf-Life and Stability Testing, Food Preservation Techniques, Chemical Residue Detection, Food Emulsion and Gelation, Food Toxicology Testing, pH and Acidity Measurement, TSS Measurement, Vitamin Content Analysis, Mineral Content Analysis, Total Dietary Fiber Content, Heavy Metal Testing, Alcohol Content Testing.

- **Chemistry Testing and Analytical Skills:**

  Titration, pH Measurement, Gravimetric Analysis, Precipitation Reaction, Flame Test, Electrochemical Analysis, Colorimetry, Kjeldahl Method, Atomic Absorption Spectroscopy, Sol-gel Method,

- **Microbial Analysis skill:**

  Gram Staining, Growth Media Culturing, Selective Media Culturing, Microscopic Examination, Total Plate Count, Total Viable Count, Most Probable Number Method, Antimicrobial Susceptibility Testing, Biochemical Test – (Catalase, Oxidase, Coagulase, Indole, Methyl Red, Urease, Voges-Proskauer (VP), Citrate Utilization, Carbohydrate Fermentation, Hydrogen Sulfide ($H_2S$) Production, Nitrate Reduction, Decarboxylation,

Methyl Red- Voges-Proskauer (MR-VP), Phenylalanine Deaminase, Urease Hydrolysis , Motility , Lactose Fermentation Test), Microbial Antioxidant Assay – (DPPH, ABTS, SOD Activity, Lipid Peroxidation Assay, Catalase Activity, Reactive Oxygen Species, Hydrogen Peroxide Scavenging), Polymerase Chain Reaction (PCR)

**Experimental Tools:**

**Nanostructured Thin Film Deposition Method:**

- **Spray Pyrolysis (Practical)** used to develop $TiO_2$ metal-doped nanostructured thin films
- **Spin Coating (Theoretical)** used to develop $TiO_2$ metal-doped nanostructured thin films

**Nanomaterials Analysis and Characterization:**

- **XRD (Theoretical)** for Crystallographic Analysis
- **TEM (Theoretical)** for Morphology, Internal Structure and Size of Nanoparticles
- **Raman Spectroscopy (Theoretical)** for Structural Features of Nanomaterials
- **FTIR (Theoretical)** for identifying Chemical Bonds and Functional Group
- **SEM (Theoretical)** imaging for Surface Morphology and Topography Analysis

**Analytical Tools:**

- **HPLC (Practical)** used to separate, identify, and quantify compounds in a mixture
- **GC (Theoretical)** used to separate and analyze volatile compounds
- **GC-MS (Theoretical)** used to identify volatile compounds
- **UV-Vis (Practical)** used to quantify compounds
- **EDXRF (Practical)** used to elemental analysis

**Dry Lab tools:**

- Chemical Drawing and Visualization: ChemDraw (Intermediate) and Biovia Draw (Intermediate)
- Quantum Chemistry: Gaussian (Intermediate)
- Virtual Screening Tool: PyRx (Intermediate)
- Molecular Docking: AutoDock (Intermediate), AutoDock Vina (Intermediate)
- Molecular Dynamics: NAMD (Beginner)
- Visualization of Biological Networks: Cytoscape (Intermediate)
- ADMET Prediction: SwissADME, ADMETlab
- Simulation: COMSOL (Beginner), VAMPIRE (Beginner)

**Staistical Analysis:** Python (NumPy, Pandas), SPSS, MiniTab

**Graphical Visualization:** Origin Pro, Python (Seaborn, Matplotlib)

**Referencing Tools:** EndNote, Mendeley.

**Programming Skills:** Python (Intermediate), R (Beginner), SQL (Intermediate)

**Machine Learning Algorithms**:

- Supervised learning (linear regression, decision trees, random forests),
- Unsupervised learning (K-means, PCA)

**Research Competencies:**

Literature Review, Experimental Design, Research Proposal, Grant Writing, Statistical Analysis, Critical Thinking, Interdisciplinary Collaboration, Data Visualization, and Presentation Skills.

**Soft Skills in Research:** Data Mining, Data Management, Data Interpretation, Proofreading, Project Writing, Paper Writing.

**OS & Tools:** Windows, Git/Github

## Language Skills:

Bengali (Native),

English *(LISTENING C1, READING B2, WRITING B2, SPOKEN PRODUCTION C1, SPOKEN INTERACTION C1, Levels: A1 and A2: Basic user; B1and B2: Independent user; C1 and C2: Proficient user)*

## Scientific Activities:

**Ongoing Work:**

1. **Akib AH**, Saha I. Synthesis of metal-doped $TiO_2$ nanostructured thin film for glucose and gas sensing applications
2. **Akib AH**, Junaid M. Parthenolide: A Review of Sources, Pharmacology, Pharmacokinetics and Toxicity

**Computational Research Projects:**

1. **Hg adsorption on gold surfaces with machine-learned interatomic potentials** (Python, ASE, MACE-MP-0, CHGNet, Linux/WSL, GitHub). Screened Hg adsorption on all 11 high-symmetry sites of Au(111), Au(100) and Au(110), a benchmark for copernicium (Cn) on the gold thermochromatography detectors used at FLNR, JINR, with convergence tests on slab size, vacuum and lattice constant. CHGNet brackets the experimental adsorption energy (≈ −98 kJ/mol), and the open facets bind Hg 14–22 kJ/mol more strongly than Au(111). github.com/aszad-akib/hg-on-gold-mlip
2. **Pd adsorption on gold: benchmarking six interatomic potentials** (Python, ASE, EAM, MACE, CHGNet, Linux/WSL, GitHub). Benchmarked Lennard-Jones, EMT, EAM, MACE-MP-0, MACE-MPA-0 and CHGNet against DFT for Pd on 11 sites of three gold facets, a model for darmstadtium (Ds) on gold detectors. The EAM alloy potential

matched DFT within 0.11 eV, and 5 of 6 potentials predict that Pd alloys into the gold surface instead of staying on top. github.com/aszad-akib/pd-on-gold-potentials

**Summer School**

- **Beihang International Summer School**
  - Nature-Inspired Materials and Chemistry | 13th July – 27th July, 2026
  - 2nd International Hangzhou Nanoelectronics Summer School | 21st July – 23rd July, 2026
- **4th ML/AI for Electron Microscopy School (Online) | University of Tennessee, Knoxville | 22th June – 26th June, 2026**
  - Project:
- **ReMag Summer School: "Recycling of Permanent Magnets and Critical Raw Materials" (Online) | EIT RawMaterials | 6th May, 2026**
- **InPlasTwin Summer School 2026 (Online) | Jožef Stefan Institute | 15th June – 19th June, 2026**
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  - Analytical Approaches for the Detection of Micro- andNanoplastics in the Environment and Food
- **Experimental Microfluidics: A Short Course (Online) | ETH Zurich | 15th June – 18th June, 2026**
- **SWEB Hybrid Summer School "Functional Coating and X-Chromic Materials (Online) | University of Latvia | 18th June – 19th June, 2026**
- **MemrisTec Summer School 2026 (Online) | 30th August – 7th September,2026**

**Summit:**

- **Nano Summit Berlin | 20th May, 2026**

**Winter School**

- **13th Virtual Winter School on Computational Chemistry (To be attended)**

**Hackathon**

- **Third Machine Learning for Microscopy Hackathon (Online) (To be participated) | University of Tennessee, Knoxville | 15th December – 17th December, 2026**

**Workshop:**

- **Computer Simulation and Theory of Macromolecules 2026 (Online) | Hünfeld | 24th - 25th April, 2026**
- **Visualizing data for scientific discovery | Royal Society of Chemistry (RSC) | 28th -29th April, 2026**

**Virtual Internship Capsule:**

- **Fusion of Advanced Functional Materials & 3D printing for Sustainability | 10 Sep 2026**

**Webinar:**

- Methods in bioinformatics (Recorded):
  Organized by EMBL, which covered a wide range of topics in Bioinformatics including Exploring Gene Ontology, Methods for exploring newly-annotated species, mass spectrometry-based proteomics, methods in bioimage analysis, genomic variant calling, rare-variant association analysis, building and analyzing biological networks and approaches for using protein-protein interaction networks for biological discovery.
- **Scaling Water Resilience and Circular Water Solutions from EU projects | Joint event EU Mission Adaptation and GD-SO | 3rd September, 2026**

**Symposium:**

- **DrugSpace 2025 Symposium** (Online) by BioSolveIT
  Attended live sessions on key areas such as small molecule discovery, scaffold extension, and manipulation, state-of-the-art lead optimization, targeting protein intermediates, screening of ultra large compound collection, latest development in AI methods and their application
- **Machine Learning in Drug Discovery Symposium** (Online**)** by Broad Institute of MIT and Harvard
  Covering topics like AI model driven drug discovery, application physics-based simulation to drug challenging targets, biomedical discovery, foundation model for drug discovery, and targeting cell states in cancer.

**Training:**

- **9 May, 2025 – 30 May, 2025 | ABCD Laboratory, Bangladesh**
  **Computer-Aided Drug Design: Desktop to Publication** (Online)

1. 50+ hours training session covering wide range of topics divided into 6 modules
2. Module A : Researching Target/Ligand/Gate-keeper residue, Databases, Use of ML in Small molecules/Chemical Identifier/Fingerprints
3. Module B : Classical Mechanics, Quantum Chemistry Method, Force Field, Small Molecule Optimization by Gaussian, Generation of publication-quality figures
4. Module C: Different Structures of Proteins, Choosing Targets, AI in Drug Design, Uniprot Database and Protein Data Bank (PDB)
5. Module D: Molecular Docking, Visualization of protein-ligand using Chimera/Pymol/Discovery Studio/Web server, Virtual Screening
6. Module E: Drug Likeness ADME/T Analysis, online and offline tools for Drug Likeness and ADME/T Analysis

7. Module F: Molecular Dynamics Simulation, NAMD, VMD

- **5 November. 2024 – 4 December, 2024 | TICI**
  **"Quality Control and Unit Operation in Food Industry"** under the ACES Department

1. 110+ hours theory and practical training course applicable for the food industry.
2. Analytical Chemistry & Environmental Sciences (76 hours):
3. Quality Control: QC and QA
4. Industrial application: Viscosity of oil.
5. Analysis of Industrial water: Nanotechnology in water treatment, Titrimetric analysis, Acidity, Alkalinity, Hardness.
6. Quality Testing methods in food industry: Antioxidant, Ascorbic Acid, Pectin, Fat, Preservatives (E-201), Adulteration, Polyphenol, Acid number, Saponification, Peroxide value, Caffeine, Total Sugar, Rancidity, Allyl isothiocyanate, Polarimeter.
7. Spectrometric methods of analysis: Spectrophotometer, Flame Photometer, UV/Vis Spectrophotometer, ICP-OES, EDXRF, FTIR, Estimation of Vitamin-A
8. Chromatographic method of analysis: GC, GC-MS, and HPLC
9. Pollution control in the food industry: DO, BOD, and COD
10. Operation & Process Technology (18 hours): Process symbol, Process diagram, ion exchange, SCADA, PID, Boiler and Heat Exchanger, Solid Handling and Absorption, Evaporation and Crystallization.
11. Instrumentation & Control Engineering (04 hours): Process control device, control loop, PCL Mechanical Engineering (04 hours): Refrigeration system, Pipe fittings and valves
12. Industrial Safety and Health (06 hours): Safety measures, Fire hazards, and fire protection management.
13. Others (04 hours): National Integrity Strategy.

- **10 January, 2025 – 12 January, 2025 | Alcumus Bangladesh Ltd.**
  **GMP and HACCP-Based Food Safety Management Systems.**

1. 20+ hours of professional training in GMP and HACCP
2. Codex Alimentarius Commission, IPPC, OIE, ISO, FSSC, IFS, GAA, GFSI
3. Food Safety Act, Rules and Regulations.
4. Putting HACCP steps and HACCP principles into action.
5. ISO/TS 22002-1, ISO/TS 22002-2, ISO/TS 22002-3, ISO/TS 22002-4, ISO/TS 22002-5, ISO/TS 22002-6,
6. Principles of Food Hygiene CAC/RCP 1-1969, Rev. 2022
7. PRP, OPRP and CCP
8. Food Safety Audit Methodology, Audit Reporting, Food Auditor's Code of Conduct and Responsibilities.

- **29 December, 2024 – 2 January, 2025 | Rural Development Academy, Bogura**
  **Agro-Processing, Preservation and Marketing**

1. 40+ hours of Profession training in GMP and HACCP

2. Yogurt Production Via Fermenting Milk with LAB
3. Rajmohon Sweet made with condensed milk
4. Patties filled with savory ingredients (e.g. meat)
5. Mixed fruit chutney made with fruits is a tangy, sweet condiment
6. Donut from fresh baked dough
7. Packaging of Ghee, Olive Pickles, and Yogurt.

**Courses:**

- **IBM Data Science Professional Certificate**
  Completed 12 courses covering topics like SQl, Python, Exploratory Data Analysis, Machine Learning, Applied Data Science, Generative AI.

- **Data Science & Machine Learning with Python for Beginners** by **Ostad**

  20 modules consisting of 60 hours of training video and assignments, quiz, and live coding test on Data Visualization, EDA and ML-based algorithms (Logistic regression, KNN, Decision Tree, Support Vector Machine, K Means Clustering, Time Series Analysis
- Other Certificates:

1. https://coursera.org/share/f04f451df17cde351f38652179e0d9ac (University of Colorado Boulder)
2. Python Programming Fundamentals (Microsoft)

**Internship:**

- **10 August – 9 September | UMT Mobility Program**

1. 150+ hours of theoretical class work and practical lab work on new product development supervised by Dr. Asma' Ali.
2. Formulated multiple prototypes and conducted Sensory Evaluation (Preference Test).
3. Performed proximate analysis Moisture (Oven Drying), Ash (Ashing), Crude Protein (Kjeldahl Method), Crude Fat (Soxhlet Method), Crude Fibre (Gravimetric Method).
4. Designed convenient packaging and led Marketing Efforts.
5. Secured Bronze on the product launch day and got praised for a unique idea related to waste utilization.

- **16 January – 15 February | Internship at BCSIR**

1. 160+ hours of lab work in Food Toxicology Research Section under Institute of Food Science and Technology.
2. Prepared Mobile Phase for Caffeine detection and Aflatoxin Detection.
3. Extracted Aflatoxin using QuCheERS method.
4. Collected beverage samples like Coffee, Cola drinks for Caffeine detection and Corn, Maize, Chanachur for Aflatoxin detection.
5. Analyzed the presence of Caffeine by HPLC using UV-Vis detector and Aflatoxin by HPLC using FLD detector.

## Tutorial:

- Preparing an analytic dataset from NHANES, and working with subset to get proper SE and 95% CI using R and Stata by Ehsan Karim
- Staistics 787, Statistics 726, Staistics 120 by Ross Ihaka
- AlphaFold A practical guide
- A Guide to UniProt for students
- Radiation Safety by Gennadiy Nikolaevich Timoshenko, JINR
- Lecture on the problems of the radiation barrier during interplanetary flights and the search for traces of life in space by Aleksander N. Bugay, JINR

## Leadership & Extra-Curricular activities:

**Scientific Organization Involvement**

- American Society for Microbiology | Student Member | 22 Nov, 2025 - Ongoing
- Material Research Society | Student Member | 1 Dec, 2025 - Ongoing
- Institute of Food Science and Technology | Student Member | 5 Dec, 2025 - Ongoing

**Leadership Roles:**

**Fitness Instructor:**

- Offer personalized coaching on effective fat loss strategies and muscle gain, tailoring advice to individual goals and needs.

**Social Work & Volunteerig:**

**Badhon**

- Member (CVASU Unit):
  Donated Blood 10 times, helped manage blood for 5 emergency patients.

## References:

1) Dr. Indrajit Saha
   Professor, Faculty of Food Science and Technology
   Chattogram Veterinary and Animal Sciences University
   Academis Supervisor for my Master's thesis
   **indrajit468@yahoo.com**
   Mr. Saha is highly experienced in semiconductor spintronics

2) Dr. Mohd Nizam Lani
   Assoc. Professor, Faculty of Food Science and Agrotechnology
   Universiti Malaysia Terengganu
   Internship co-ordinator in **UMT Mobility Program**
   **nizamlani@umt.edu.my**

Mr. Lani is highly experienced in Food Microbiology & Safety

3) Md. Junaid
   Founder and Principal Investigator
   ABCD Laboratory, Bangladesh
   Research Supervisor in ABCD lab
   junaidresearchgroup1@gmail.com
   Mr. Junaid specializes in AI driven drug discovery
4) Dr. Asma' Binti Ali
   Assoc. Professor, Faculty of Food Science and Agrotechnology
   Universiti Malaysia Terengganu
   Internship supervisor in **UMT Mobility Program**
   asma.ali@umt.edu.my